

Core Signatures and Inversions

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Abstract

Signatures provide a feature representation for paths that respects concatenation through Chen’s identity, but the truncated signature is redundant: it carries the same information as the much smaller log signature. We introduce *core signatures*, a subset of the truncated signature with the same dimension as the log signature and no change of basis required — the core entries are read directly from the signature, and the remaining entries are recovered through closed-form substitution rules. Structurally, the core entries are the Lyndon coordinates of the step- n free nilpotent Lie group; this identification is classical (Reutenauer 1993; Amendola–Friz–Sturm 2019) but the algorithmic packaging is new. We then use Chen’s identity to invert a core signature into a piecewise-linear path of two, three, or more segments that exactly matches the target signature up to a chosen truncation level. For more than three segments the inversion takes the form of a weighted nonlinear least-squares regression in signature space, and we observe (and conjecture rigorously) that this regression coincides with the optimal L^1 piecewise-linear approximant of the target. The accompanying Mathematica code is at <https://github.com/MarcosCarreira/CoreSignatures>; an application to USD Treasury and Brazilian DI futures curves is in preparation.

Keywords: signatures; log signatures; Lyndon words; signature inversion; term-structure modelling

1 Introduction

Term structures — interest rates, implied volatilities, futures curves — are functions of time-to-maturity that practitioners want to compress into a small number of geometric features. Principal components do this in the linear regime; signatures generalise the idea in a way that respects concatenation. For a path $X : [a, b] \rightarrow \mathbb{R}^d$, the signature $S(X)_{a,b}$ collects all iterated integrals along X into a single tensor series; truncated at level m , it provides a finite-dimensional encoding of X that is faithful up to tree-like equivalence. The truncated signature contains redundancies coming from the shuffle relations, and the standard remedy is the log signature — whose dimension is that of the free nilpotent Lie algebra $\mathfrak{g}^m(\mathbb{R}^d)$, typically computed in practice through Reizenstein’s `iisignature` library [Reizenstein (2018), Reizenstein and Graham (2020), Reizenstein (2015)].

This paper develops two results aimed at practitioners working with truncated signatures. First, we identify a subset of the truncated signature — the *core signature* — that has the same dimension as the log signature but consists of actual signature entries. The core entries are exactly the entries indexed by Lyndon words on the alphabet $\{1, \dots, d\}$, their count is given by Witt’s formula, and the remaining entries are recovered through explicit substitution rules obtained from the shuffle relations. Second, we use Chen’s identity to invert a core signature into a concatenation of two or three line segments in the time-ordered case — the case relevant for term structures — and into longer concatenations through a weighted nonlinear least-squares regression in signature space. We give numerical evidence that this regression coincides with the

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optimal L^1 piecewise-linear approximant and state the precise norm-equivalence conjecture that would make the coincidence rigorous.

The structural identification of core entries with Lyndon coordinates follows from results that are classical in the algebraic-combinatorics literature ([Reutenauer (1993)], the Lyndon basis theorem and Radford's theorem) and in the rough-paths literature ([Amendola, Friz and Sturmfels (2019)]), the universal variety $G^n(\mathbb{R}^d)$. Our contribution is therefore computational packaging — naming the subset, exhibiting an explicit shuffle-product elimination that produces it, tabulating the substitution rules Q_m , and using these coordinates for Chen-based inversion — rather than a new coordinate theorem. For previous signature-inversion algorithms see [Fermanian et al. (2023)].

After our 2023 preprint, [Amendola et al. (2025)] introduced a Macaulay2 package (`PathSignatures`) whose `lyndonShuffle` method implements the same Lyndon-shuffle decomposition that underlies our substitution rules Q_m . In their Section 6 they use the term *core signature tensor* with a different meaning: the signature of a base path that generates a family by linear transformation, suited to the algebraic geometry of signature varieties rather than to Chen-based inversion of individual signatures. The terminology overlap is unfortunate but the constructions are distinct. A parallel Julia/OSCAR implementation is in [Riffo and Schmitz (2026)].

2 Signatures

A path is a continuous map $X : [a, b] \rightarrow \mathbb{R}^d$ of bounded variation; we reparametrise to $[0, 1]$ when convenient. The signature truncated at level m is

$$S^{(\leq m)}(X)_{a,b} = \left(1, S^{i_1}, S^{i_1, i_2}, \dots, S^{i_1, \dots, i_m} \right), \quad i_k \in \{1, \dots, d\}, \quad (1)$$

with level- k entries

$$S^{i_1, \dots, i_k} = \int_a^b \int_a^{t_k} \dots \int_a^{t_2} dX_{t_1}^{i_1} \dots dX_{t_k}^{i_k}. \quad (2)$$

For a single line segment with slopes $(\beta_1, \dots, \beta_d)$ over an interval of length T , every entry has a closed form:

$$S^{i_1, \dots, i_k} = \frac{\beta_{i_1} \dots \beta_{i_k} T^k}{k!}. \quad (3)$$

This is the building block we will use throughout the inversion.

Shuffle product. A (p, q) -shuffle of two ordered sequences of lengths p, q is an interleaving that preserves the relative order within each sequence; $I \sqcup J$ denotes the multiset of all shuffles of multi-indices I and J . The fundamental shuffle identity — valid for any path — is

$$S^I \cdot S^J = \sum_{K \in I \sqcup J} S^K. \quad (4)$$

Equation (4) is the algebraic engine of Section 3: it provides the relations that reduce the truncated signature to its core.

Chen's identity. For consecutive paths $X : [a, b] \rightarrow \mathbb{R}^d$ and $Y : [b, c] \rightarrow \mathbb{R}^d$, the concatenation $Z = X * Y$ has signature

$$S(Z)_{a,c} = S(X)_{a,b} \otimes S(Y)_{b,c}, \quad \text{or componentwise} \quad z_{i_1, \dots, i_k} = \sum_{j=0}^k x_{i_1, \dots, i_j} \cdot y_{i_{j+1}, \dots, i_k}, \quad (5)$$

with the conventions $x_\emptyset = y_\emptyset = 1$. The componentwise form allows us to compute individual entries of Z without rebuilding the full concatenated signature — the basis of the inversion algorithm in Section 4.

3 Core Signatures

The shuffle relations mean that not every entry of the truncated signature carries new information. The standard parsimonious encoding is the log signature, but it requires a change of basis (specifically, a Hall-basis projection of the Lie series). We show that a subset of the truncated signature itself — the *core elements* — has the same dimension and the same information content as the log signature, can be read directly from $S(X)$, and admits an explicit structural characterisation in terms of Lyndon words.

3.1 Core variables via the shuffle product

Let $V_m = \{s_w : |w| = m\}$ denote the set of signature entries at level m , and let C_m be the *core set up to level m* , defined inductively as the entries that cannot be expressed in terms of lower-indexed and lower-level entries via the shuffle relations. For $d = 2$, the recursive application of (4) at level 3 produces the Gröbner basis

$$\begin{aligned} & 3s_{1,1,1} - s_1s_{1,1}, \quad -s_2s_{1,1} + s_1s_{2,1} + s_{1,1,2} - s_{2,1,1}, \\ & -s_1s_{2,1} + s_{1,2,1} + 2s_{2,1,1}, \quad -s_2s_{1,2} + s_2s_{2,1} + 2s_{1,2,2} - 2s_{2,2,1}, \\ & -s_2s_{2,1} + s_{2,1,2} + 2s_{2,2,1}, \quad 3s_{2,2,2} - s_2s_{2,2}. \end{aligned}$$

After eliminating six variables in terms of $C_2 = \{s_1, s_2, s_{1,2}\}$ and the lex-minimal representatives of each remaining orbit, two entries remain free:

$$C_3 = C_2 \cup \{s_{1,1,2}, s_{1,2,2}\}.$$

Repeating the construction at higher levels yields the core sizes for $d = 2$:

$$|C_m| \text{ for } m = 1, \dots, 6 = 2, 3, 5, 8, 14, 23, \tag{6}$$

matching the log-signature column of Reizenstein’s tables. For $d = 3$ we obtain 3, 6, 14, 32, 80, 196, again matching the log signature. The Lyndon multi-indices that constitute C_4, C_5, C_6 are listed in Appendix A; the substitution rules Q_m for $m \leq 5$ are tabulated there in full, and Q_6 for $d \in \{2, 3\}$ is exported in the companion notebook `CoreSignatures202412.nb`.

m	$d = 2$ sig	$d = 2$ core/log	$d = 3$ sig	$d = 3$ core/log	$d = 4$ sig	$d = 4$ core/log
1	2	2	3	3	4	4
2	6	3	12	6	20	10
3	14	5	39	14	84	30
4	30	8	120	32	340	90
5	62	14	363	80	1364	294
6	126	23	1092	196	5460	964

Table 1: Sizes of signatures and core signatures at low levels. Core sizes coincide with the log-signature sizes of [Reizenstein (2015)].

3.2 Core elements as Lyndon coordinates

The numerical coincidence $|C_m| = \dim \mathfrak{g}^m(\mathbb{R}^d)$ and the choice of the lex-minimal representative in each shuffle orbit are consequences of the Lyndon basis of the free Lie algebra. Fix the lexicographic order on $\mathcal{A} = \{1, \dots, d\}$; a non-empty word w is a *Lyndon word* if $w < vu$ for every non-trivial factorisation $w = uv$. For $d = 2$, the Lyndon words of length ≤ 6 on $\{1, 2\}$ are listed in Table 2; comparison with C_6 shows the multi-indices agree exactly.

Length	Lyndon words on $\{1, 2\}$
1	1, 2
2	12
3	112, 122
4	1112, 1122, 1222
5	11112, 11122, 11212, 11222, 12122, 12222
6	111112, 111122, 111212, 111222, 112122, 112212, 112222, 121222, 122222

Table 2: Lyndon words on $\{1, 2\}$ up to length 6. These are exactly the multi-indices of the core variables.

Proposition 1 (Core variables are Lyndon coordinates). *The core sets C_m identified by the shuffle-product computation are exactly the Lyndon words on $\{1, \dots, d\}$ of length $\leq m$. The corresponding σ_w form a transcendence basis for the coordinate ring of the universal variety $G^m(\mathbb{R}^d)$, and the substitution rules Q_m are the explicit form of the corresponding Radford isomorphism truncated at level m .*

The proof uses three classical ingredients: the Lyndon basis theorem for the free Lie algebra ([Reutenauer (1993)] Theorem 5.1), Poincaré–Birkhoff–Witt, and Radford’s theorem that the shuffle algebra is the free commutative algebra on Lyndon words ([Reutenauer (1993)] Theorem 6.1). Together they identify the Lyndon-indexed signature entries with algebraically independent coordinates on the universal variety $G^n(\mathbb{R}^d)$ studied in [Amendola, Friz and Sturmfels (2019)]; the lex-minimal choice within each shuffle orbit corresponds to the Lyndon representative by Duval’s factorisation theorem ([Reutenauer (1993)] Theorem 5.1.5). The full derivation is given in the companion arXiv preprint.

3.3 Witt formula and the Pascal-triangle structure

The number of Lyndon words of length n on a d -letter alphabet is given by Witt’s formula ([Reutenauer (1993)] Section 4):

$$\ell_d(n) = \frac{1}{n} \sum_{k|n} \mu\left(\frac{n}{k}\right) d^k, \quad (7)$$

where μ is the Möbius function. For $d = 2$ this gives the sequence 2, 1, 2, 3, 6, 9, 18, 30, $\dots$ whose cumulative sums match (6). The multi-graded refinement of (7) — counting Lyndon words on $\{1, 2\}$ with prescribed numbers of 1s and 2s — recovers the Pascal-triangle distribution we observed empirically in the substitution rules.

4 Inversion of Core Signatures

We use Chen’s identity (5) to recover, from a target core signature, a piecewise-linear path whose own core signature exactly matches the target up to a chosen truncation level. We treat the time-ordered case $X(t) = (t, f(t))$ in detail — this is the case relevant for term structures — and outline the general 2d case at the end.

Related work on the inversion problem. Three recent papers from the Berlin/TU group develop the inversion problem from an algebraic-geometric angle that complements ours. [Amendola, Lotter and Schmitz (2026)] generalise the piecewise-linear and polynomial classes of [Amendola, Friz and Sturmfels (2019)] to piecewise polynomial paths and splines, and introduce the *path recovery degree* (Definition 6.1) as the number of complex points in a generic fibre of the signature map: the formal counterpart of our empirical observation that the two-piece

time-ordered system has two solutions. For $d = 2$ at level $k = 4$, their Table 3 records the recovery degree as 4 for piecewise linear paths with four segments and up to 40 for higher-degree spline classes. [Schmitz (2025)] gives an $O(d^4)$ algorithm for the level-3 d -segment piecewise linear case using Gauss transformations on signature tensors as a tensor analogue of congruence normal forms, outperforming Gröbner-basis methods. [Amendola and Schmitz (2025)] treats the adjacent problem of path recovery from *barycenters* of multiple signatures, relevant when the input is an expected signature.

We develop the time-ordered case $X(t) = (t, f(t))$ in detail because the algebra closes with closed-form Chen-identity solutions for two and three segments (Appendix A), and because this is the case most relevant for term structures; the same machinery extends to general two-dimensional paths — see the semicircle and digit-5 examples below — and in principle to higher dimensions. For multi-piece inversion we use a weighted regression in signature space whose moment-matching interpretation, via the Lyndon coordinates of Section 3, links signature-space minimisation to classical L^1 approximation theory — a connection that is, to our knowledge, not present in the algebraic-geometric line of work above.

4.1 Time-ordered paths: two pieces

Let $X(t) = (t, f(t))$ on $[0, T]$. We seek a concatenation $X * Y$ of two line segments

$$X(t)|_{[0, t_1]} = (t, \beta_1 t), \quad Y(t)|_{[t_1, T]} = (t, \beta_1 T + \beta_2 t),$$

parametrised by $\{T, t_1, \beta_1, \beta_2\}$, whose core signature matches a target $\{z_1, z_2, z_{1,2}, z_{1,1,2}, z_{1,2,2}\}$. By (3) and (5), the matching conditions are the five equations of Table 3.

Level	Eq #	Equation
1	1	$T = z_1$
1	2	$\beta_1 t_1 + \beta_2 (T - t_1) = z_2$
2	3	$\frac{1}{2}\beta_1 t_1^2 + \frac{1}{2}\beta_2 (T - t_1)^2 + \beta_2 t_1 (T - t_1) = z_{1,2}$
3	4	$\frac{1}{6}\beta_1 t_1^3 + \frac{1}{6}\beta_2 (T - t_1)^3 + \frac{1}{2}\beta_2 t_1 (T - t_1)^2 + \frac{1}{2}\beta_2 t_1^2 (T - t_1) = z_{1,1,2}$
3	5	$\frac{1}{6}\beta_1^2 t_1^3 + \frac{1}{6}\beta_2^2 (T - t_1)^3 + \frac{1}{2}\beta_2^2 t_1 (T - t_1)^2 + \frac{1}{2}\beta_1 \beta_2 t_1^2 (T - t_1) = z_{1,2,2}$

Table 3: System for the two-piece time-ordered inversion.

The system has five equations in four unknowns and admits two solutions, obtained by dropping equation #5 or equation #4; the closed forms are in the companion notebook, and substituting either solution into the dropped equation gives the consistency condition

$$2z_{1,2}^2 + z_1 z_2 z_{1,2} = 3z_1 z_{1,2,2} + 3z_2 z_{1,1,2}, \quad (8)$$

which expresses the algebraic limit of two segments.

Worked examples. For $f(t) = \sin(\pi t)$ on $[0, 1]$, the target core signature is $\{1, 0, -2/\pi, -1/\pi, 1/4\}$. Solution 1 (dropping equation #5) gives the symmetric tent with peak $(1/2, 4/\pi)$. For $f(t) = \sin(\pi t^2)$ Solution 1 gives an asymmetric tent with peak slightly past $t = 1/2$, matching the geometric asymmetry of the curve (Figure 1). For $f(t) = \sin(\pi t^2) + 1 - t$ both solutions exist and produce two two-segment fits whose crossing point sits very close to the original curve (Figure 2); the two values of t_1 bracket the location of maximum curvature.

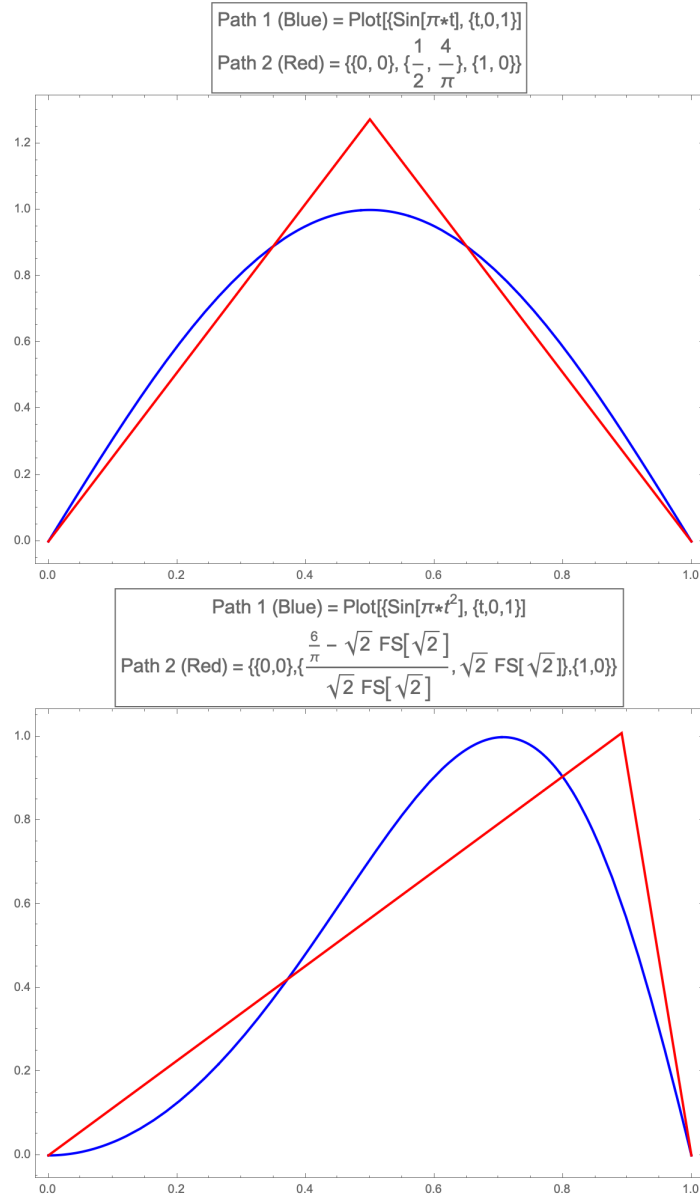


Figure 1: Two-segment fits to $\{t, \sin(\pi t)\}$ (left) and $\{t, \sin(\pi t^2)\}$ (right).

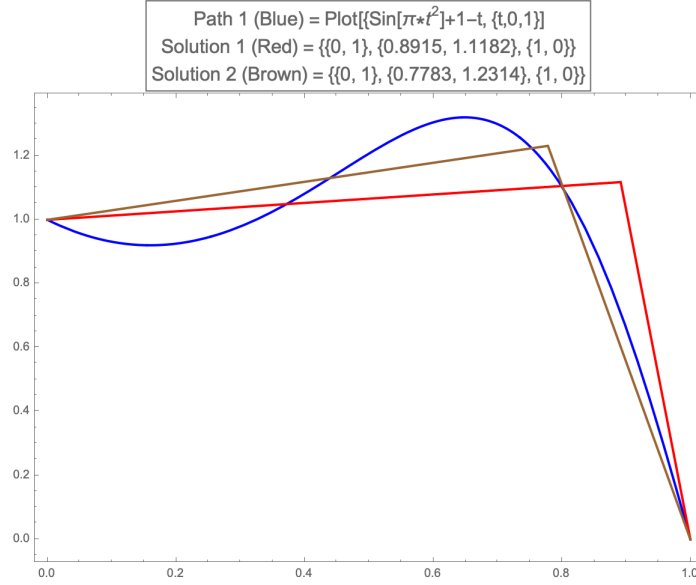


Figure 2: The two two-segment fits to $\{t, \sin(\pi t^2) + 1 - t\}$.

4.2 Adversarial cases and three pieces

The two-piece inversion fails when $z_1 z_2 - 2z_{1,2} = 0$ (Lévy area zero relative to the chord) or when $z_2 = 0$ removes one of the two solutions. Both situations occur for $\{t, \sin(2\pi t)\}$ and similar oscillatory targets. The remedy is to add a third segment; the extended system has three more unknowns and four more equations from level 4 and the cross-terms (full listing in the companion notebook, section **ThreePiece**), with the same overdetermined-Gröbner structure and its own closed-form solutions for each choice of dropped equation.

4.3 Four or more pieces, and the regression formulation

For four or more segments the algebraic Gröbner approach becomes prohibitively expensive. The practical alternative is a nonlinear weighted least-squares regression in signature space: with θ parameterising the piecewise-linear path and $\mathcal{L}_m$ the set of Lyndon words up to level m ,

$$\theta^* = \arg \min_{\theta} \sum_{w \in \mathcal{L}_m} w_w (\sigma_w[g_\theta] - \sigma_w[X])^2. \quad (9)$$

The companion notebook uses level-only weights $w_w = W_{|w|}$ with $W_\bullet = \{50, 50, 30, 10, 10, 3, 3, 3\}$ across the core indices and **DifferentialEvolution** as the optimiser. Signature entries at level k on a path of duration T scale as $O(T^k/k!)$ via the simplex of integration, so the weights are decreasing with level to keep all residuals on comparable footing.

Figures 3–4 show four- and five-segment fits to oscillatory time-ordered paths, to the semicircle, and to the digit “5”. The five-piece digit-5 fit recovers the topology of the original 7-segment path to visual accuracy.

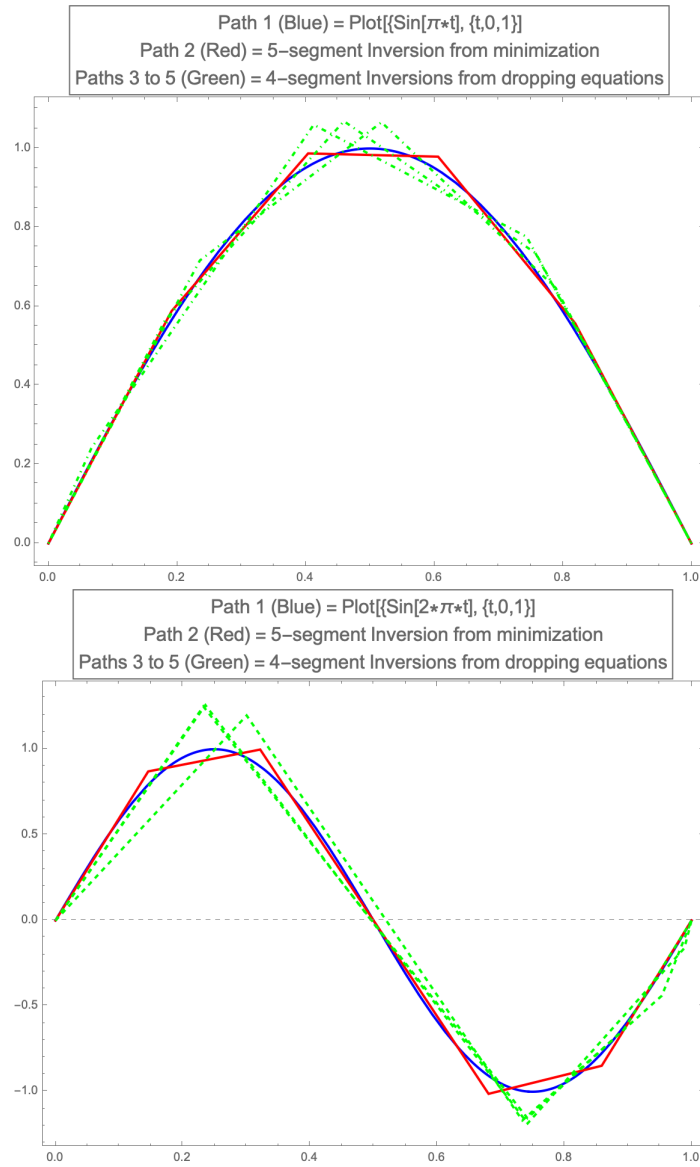


Figure 3: Fits of 4 and 5 segments to $\{t, \sin(k\pi t)\}$.

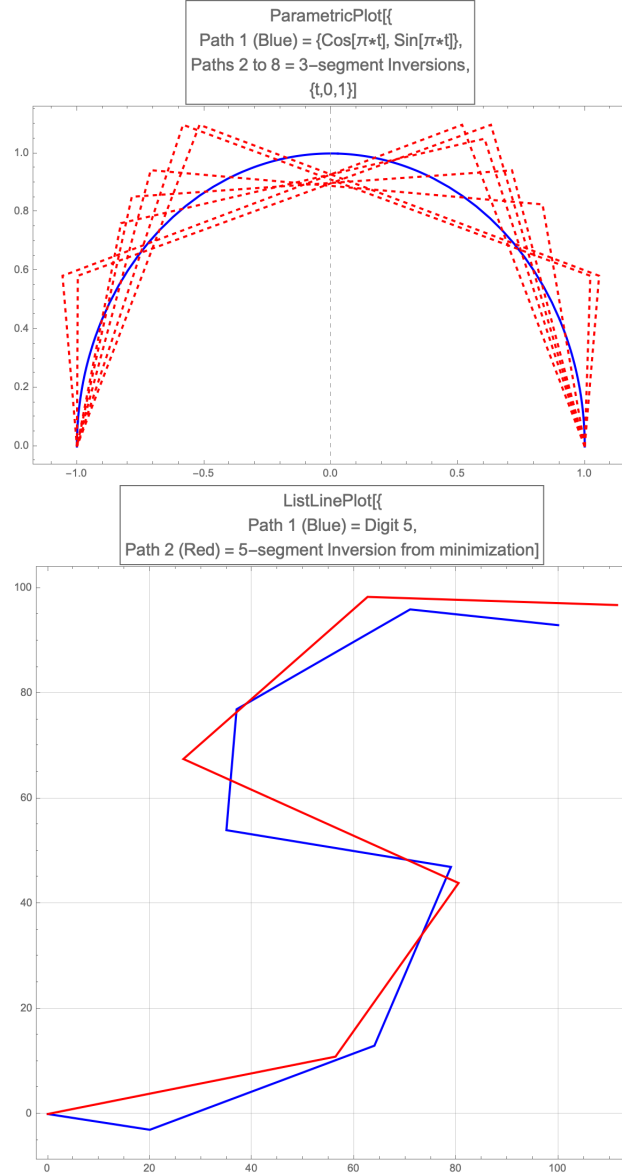


Figure 4: Three-piece inversion of the semicircle (left) and five-segment inversion of the digit 5 (right).

4.4 Signature entries as iterated path moments

For $X(t) = (t, f(t))$ on $[0, T]$, integration by parts gives

$$\sigma_{1,2}[X] = Tf(T) - \int_0^T f(t) dt, \quad \sigma_{1,1,2}[X] = \frac{T^2}{2} f(T) - \int_0^T tf(t) dt,$$

and analogously for higher Lyndon entries. To leading order in the level grading, every Lyndon-word signature entry of a time-augmented path is a polynomial in the moments $M_k(f) = \int_0^T t^k f(t) dt$ and the boundary values. The regression (9) is therefore, asymptotically, a weighted moment-matching of g_θ to f . The classical Hausdorff moment problem ([Talenti (1987)]) then tells us that matching the first m moments of $f \in C^s([0, T])$ pins down f in L^1 to order $O(m^{-s})$, which is the same rate one would get from optimal piecewise-polynomial approximation.

We can give that observation quantitative form. The companion notebook **Frechet.nb** runs an evolutionary algorithm on the four-segment piecewise-linear class minimising the mean

pointwise distance¹

$$\text{meandist}(g_\theta, f) = \frac{1}{N+1} \sum_{i=0}^N |f(t_i) - g_\theta(t_i)|, \quad t_i = iT/N.$$

On the target $f(t) = \sin(2\pi t^2)$, the mean- L^1 evolutionary algorithm and the algebraic core-signature inversion of the preceding subsections converge to qualitatively similar piecewise-linear approximations, with interior breakpoints agreeing to within a few percent. The two algorithms minimise different objectives — signature-space residual sum for the algebraic inversion, mean L^1 distance for the evolutionary algorithm — and converge to distinct nearby minima rather than to a common optimum: the multi-segment analogue of the single-segment 4.7% obstruction noted below.

4.5 The norm-equivalence conjecture

A heuristic explanation for the coincidence is visible: both methods determine the breakpoints by first-order conditions on identical leading moments of f , differing only by terms of order $\sup |f''| \cdot (\Delta t)^2$. This is an approximation-theoretic intuition rather than a proof of norm equivalence; we conjecture that the rigorous form is:

Conjecture 2 (Signature-norm equivalence). *For f, g in a class $\mathcal{F} \subseteq C^1([0, T]; \mathbb{R})$ of paths of bounded derivative, there exist weights w_w depending only on $|w|$ and T , with $w_w \sim (k!)^2/T^{2k}$ for $|w| = k$, and a Sobolev exponent $s \geq 0$ such that*

$$\sum_{w \in \mathcal{L}} w_w (\sigma_w[G] - \sigma_w[X])^2 \asymp \|f - g\|_{H^s([0, T])}^2$$

uniformly for f, g in compact subsets of $\mathcal{F}$, with constants of equivalence depending only on the diameter of the compact set in H^s .

The conjecture is supported by: (i) the empirical coincidence above on a finite test set, (ii) the moment-matching identification, and (iii) the classical link between moment matching and L^1/L^2 approximation. A direct computation on the simplest case — one free interior breakpoint (τ, y) on $[0, T]$ with pinned endpoints — shows that the two minimisers do not coincide pointwise: for $f(t) = \sin(\pi t)$, $T = 1$, symmetry forces $\tau^* = 1/2$ and the values are $y_\sigma = 4/\pi \approx 1.273$ versus $y_{L^2} = 12/\pi^2 \approx 1.216$, a 4.7% discrepancy. The conjecture is therefore not pointwise identity of minimisers but norm equivalence in the overdetermined regime, with the constants of equivalence depending on the smoothness of f . The natural first-proof target is the class of *polynomial paths*, where the universal-variety coordinates are described explicitly in [Amendola, Friz and Sturmfels (2019)] and [Pfeffer, Seigal and Sturmfels (2018)] and the moment-matching identities reduce to finite-dimensional linear systems. An empirical complement is to calibrate the weights w_w on representative training sets spanning both *rough paths* (where level- k entries can saturate the $O(T^k/k!)$ bound and the worst-case scaling $(k!)^2/T^{2k}$ is natural) and *smooth targets* (where the actual entries shrink faster and decreasing weights win on signal-to-noise); the calibrated asymptotic would identify the smoothness class on which the simple conjecture holds, versus one requiring a target-dependent weight scheme.

Concatenation. By Chen’s identity and the linearity of moments, segment-wise regression errors compose into a global error bounded by their sum: for a partition of $[0, T]$ into N pieces $X^{(i)}$ with per-segment approximations $g^{(i)}$,

$$\sum_{w \in \mathcal{L}_m} w_w (\sigma_w[g] - \sigma_w[X])^2 \leq C(m, N) \sum_{i=1}^N \sum_{w \in \mathcal{L}_m} w_w (\sigma_w[g^{(i)}] - \sigma_w[X^{(i)}])^2.$$

¹Despite the filename, the objective is not a Fréchet distance: since both paths share the time coordinate, $\text{EuclideanDistance}((t, f(t)), (t, g_\theta(t))) = |f(t) - g_\theta(t)|$, so the algorithm is minimising the mean L^1 error.

A target f of characteristic frequency ω_0 then needs $N \sim \omega_0 T$ segments of length $\Delta t \sim 1/\omega_0$ to keep the per-segment level requirement bounded; the global L^1 error scales as $O(N^{-(m+1)})$ for $f \in C^{m+1}$, matching classical piecewise-polynomial approximation rates.

4.6 Dictionary

The picture this section develops is the following correspondence between algebraic objects from Section 3 and approximation-theoretic objects.

Core-signature inversion	Approximation theory
Lyndon coordinates of $G^n(\mathbb{R}^d)$	Coordinates on the universal variety
Algebraic exact inversion (Sec. 4)	Galerkin projection in determined regime
Weighted regression (9)	Sobolev projection (under Conjecture 2)
Chen concatenation	Additive error decomposition over partition
Truncation level m	Moment-matching truncation parameter

5 Conclusions and forthcoming work

We introduced core signatures, identified them structurally with the Lyndon coordinates of the universal variety $G^n(\mathbb{R}^d)$ (Proposition 1), sized them via Witt’s formula, and gave an explicit Gröbner-basis algorithm for the substitution rules that express any signature entry in core form. We then used Chen’s identity to invert core signatures into piecewise-linear paths of two, three, or more segments, and reframed the multi-segment case as a weighted nonlinear least-squares regression in signature space whose minimiser empirically coincides with the optimal piecewise-linear L^1 approximant.

The application of the core-signature framework to interest-rate term structures is in preparation. The companion paper applies the inversion to the historical USD Treasury curve and Brazilian DI futures curve to extract a low-dimensional geometric encoding of curve dynamics suitable for stress testing, hedging, and historical-pattern recognition across the two markets. The framework has also been adopted in practitioner work on quantitative-trading future testing; see [Bloch (2024)]. We thank Sam Cohen (Oxford) for helpful discussions.

Use of AI assistance. The author used Anthropic’s Claude to review the mathematical content of the 2023 version of this paper, tighten the exposition, supply the theoretical citation chain through the Lyndon basis theorem, Poincaré–Birkhoff–Witt and Radford’s theorem that identifies the core variables with the Lyndon coordinates of $G^n(\mathbb{R}^d)$, and help draft the regression and norm-equivalence framing of Sections 4.3–4.5. All mathematical claims, the choice of which results to state as conjectures, and the final form of the paper are the author’s responsibility.

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A Substitution rules Q_m for $d = 2$

The substitution rules express every non-core signature entry as a polynomial in core entries. The cores referenced below are

$$C_1 = \{s_1, s_2\}, \quad C_2 = C_1 \cup \{s_{1,2}\}, \quad C_3 = C_2 \cup \{s_{1,1,2}, s_{1,2,2}\}, \\ C_4 = C_3 \cup \{s_{1,1,1,2}, s_{1,1,2,2}, s_{1,2,2,2}\}.$$

Level 2. $s_{1,1} = \frac{1}{2}s_1^2, \quad s_{2,1} = s_1s_2 - s_{1,2}, \quad s_{2,2} = \frac{1}{2}s_2^2.$

Level 3. $s_{1,1,1} = \frac{1}{6}s_1^3, \quad s_{1,2,1} = s_1s_{1,2} - 2s_{1,1,2}, \quad s_{2,1,1} = -s_1s_{1,2} + s_{1,1,2} + \frac{1}{2}s_2s_1^2, \quad s_{2,1,2} = s_2s_{1,2} - 2s_{1,2,2}, \quad s_{2,2,1} = -s_2s_{1,2} + s_{1,2,2} + \frac{1}{2}s_1s_2^2, \quad s_{2,2,2} = \frac{1}{6}s_2^3.$

Levels 4–6. The level-4 and level-5 substitution rules are listed in the companion notebook `CoreSignatures202412.nb`, sections Q4 and Q5; the level-6 rules for $d \in \{2, 3\}$ are in sections Q6d2 and Q6d3.